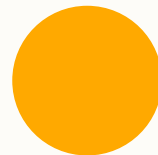




Algorithms + Pseudocode

Reese Hatfield



0



Javascript

- Long ago,
- We briefly discussed the idea of what an algorithm is.
- **A series of deterministic, correct and finite steps to solve a problem**
- We now have the ability to program





Javascript

- Javascript is Turing Complete
 - Effectively lets us implement any computable algorithm in it
 - There are better tools (usually)
 - Let's revisit algorithms with this idea in mind
- 



Javascript

- Algorithms are language agnostic
- Any given algorithm can be implemented in any programming language, not just JS
- What would an algorithm look like?
 - If not in a specific language





Javascript

- **Pseudo Code**
- A description of an algorithm at a very high level
- Can chose to implement this in any language
- Let's look at some examples





Javascript

- **Ask user how tall they are**
- **Read in data**
- **If user is taller than 6 ft**
 - **Tell them they are tall**
- **Else**
 - **Tell them they are short**





Javascript

- Simple pseudo code
- Define an algorithm
- We could implement this algorithm in any language
- Let's do it in another language
- Python
 - **You don't need to know python for this class, just an example**





Javascript

- This is a good starting place for implementing any problem
- Idea of your algorithm
- Let's try another example
- Grading application





Loops

- Grading Application
- Determine the letter grade
- Of a NUMBER of students (say 3)
- They *each* give us
 - Name
 - Percentage
- Spit out their letter grade





Loops

- Certain people, say Reese or Andrew
- Will always have an A, regardless of percentage
- A big problem
 - At least for what we've covered
 - Let's pseudo-code it





- Ask for number of students





- Ask for number of students
- For each student:
 - Ask for name
 - Ask for grade





- Ask for number of students
- For each student:
 - Ask for name
 - Ask for grade
 - Declare grade





- Ask for number of students
- For each student:
 - Ask for name
 - Ask for grade
 - Declare grade
 - If name > 90
 - Grade $\leftarrow$ A
 - Else if name > 80
 - Grade $\leftarrow$ B
 - Etc.





- Ask for number of students
- For each student:
 - Ask for name
 - Ask for grade
 - Declare grade
 - If name > 90
 - Grade $\leftarrow$ A
 - Else if name > 80
 - Grade $\leftarrow$ B
 - Etc.
 - If name is Reese or Andrew
 - Display "A"
 - Else
 - Display grade



Final Pseudo Code For Our Program

- Ask for number of students
- For each student:
 - Ask for name
 - Ask for grade
 - Declare grade
 - If name > 90
 - Grade $\leftarrow$ A
 - Else if name > 80
 - Grade $\leftarrow$ B
 - Etc.
 - If name is Reese or Andrew
 - Display "A"
 - Else
 - Display grade

